Huawei Access Node HSR-ACC-74 Routing Protocol Flap Resolution SOP

Node: HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4917

Description

HSR-ACC-74, an access node deployed at HSR Layout Access, experiences a critical routing protocol flap failure, as indicated by the ALARM_4917 code, which corresponds to a BGP route churn, OSPF adjacency loss, or ISIS LSP flooding issue. This failure is attributed to the complex interplay between Huawei's proprietary routing protocols and the node's architecture. The Access node, in this case, relies on Huawei's advanced routing protocols, such as OSPF, ISIS, and BGP, to ensure efficient routing and data exchange between the network's core and edge components. However, the intricate relationships between these protocols and the node's underlying hardware can sometimes lead to suboptimal behavior, resulting in routing protocol flaps. Furthermore, the Huawei node's architecture, which includes features such as hierarchical routing, traffic engineering, and link aggregation, can exacerbate the issue by introducing additional complexity and potential points of failure. Therefore, resolving this failure requires a careful and systematic approach that considers the node's architecture, the routing protocols in use, and the potential causes of the failure.

Pre-requisite Checks

- check: Verify the site location and ensure access to the HSR Layout Access site.
- check: Confirm the redundant link status to ensure there are no link failures affecting the node.
- check: Dispatch remote hands to the site, if necessary, to facilitate troubleshooting and repair.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfiguration or neighboring node issues.	High
OSPF adjacency loss due to network instability, neighbor issues, or misconfiguration.	Medium
ISIS LSP flooding due to network congestion, neighbor issues, or misconfiguration.	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration, Expected Output: Current configuration of the node, including routing protocols and neighbor relationships., Description: This command provides a comprehensive view of the node's configuration, allowing technicians to identify potential issues with routing protocols and neighbor relationships.

Command: display bgp summary, Expected Output: Summary of BGP neighbors, routes, and peers., Description: This command provides a high-level overview of the node's BGP neighbors, routes, and peers, helping technicians to identify issues with BGP routing.
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Command: display ospf neighbor, Expected Output: List of OSPF neighbors, including their states and adjacencies., Description: This command provides detailed information about the node's OSPF neighbors, including their states and adjacencies, allowing technicians to identify potential issues with OSPF routing.
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Resolution Steps

Step 1: Step: Verify BGP neighbor relationships and configurations., Command: display bgp neighbor, Description: This step involves verifying the BGP neighbor relationships and configurations on the node, ensuring that all neighbors are correctly configured and established.

Step 2: Step: Reset BGP peers and re-establish neighbor relationships., Command: reset bgp peer, Description: This step involves resetting the BGP peers and re-establishing the neighbor relationships, which can help resolve issues with BGP routing.

Step 3: Step: Verify OSPF neighbor relationships and configurations., Command: display ospf neighbor, Description: This step involves verifying the OSPF neighbor relationships and configurations on the node, ensuring that all neighbors are correctly configured and established.

Post-Check Verification

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Command: display bgp summary, Expected Output: Summary of BGP neighbors, routes, and peers., Description: This command provides a high-level overview of the node's BGP neighbors, routes, and peers, helping technicians to verify the resolution of BGP routing issues.

Command: display ospf neighbor, Expected Output: List of OSPF neighbors, including their states and adjacencies., Description: This command provides detailed information about the node's OSPF neighbors, including their states and adjacencies, allowing technicians to verify the resolution of OSPF routing issues.
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Roll-back Procedures

- Step: Restore the previous configuration., Command: restore configuration, Description: This step involves restoring the previous configuration, which can help recover the node from any unintended changes made during the troubleshooting process.

- Step: Re-establish BGP and OSPF neighbor relationships., Command: reset bgp peer, Description: This step involves re-establishing the BGP and OSPF neighbor relationships, which can help recover the node from any issues caused by the troubleshooting process.